

# Study Material

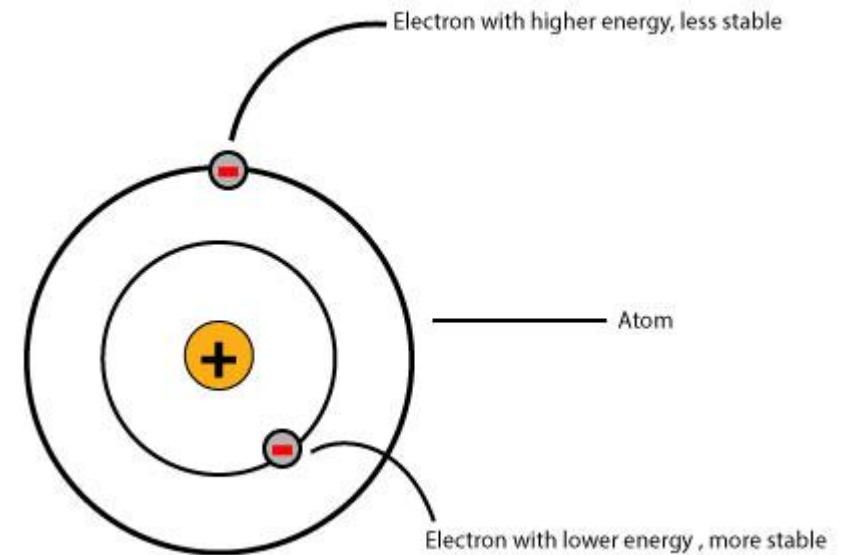
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- Electronic Principles by Albert Paul Malvino (7<sup>th</sup> Edition)
- Electronic Devices and Circuits Theory by Robert L. Boylestad (9<sup>th</sup> Ed)
- And don't forgot to browse internet for any topic
- These slides will be available after every lecture

# Introduction: Concepts of Electricity

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- Electricity is movement of **Electrons**
- **What is an Electron?**
  - a stable subatomic particle with a charge of negative electricity, found in all atoms
  - Every material has different number of electrons
- These electrons causes **charge**



# Concepts of Electricity contd.

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- Electric Charge is the property of subatomic particles that causes it to experience a force when placed in an electromagnetic field
- It can be negative or positive
- Electrons carry negative charge and Protons carry positive charge
- Three basic principles which are important in electricity are
  - Voltage
  - Current
  - Resistance

# Concepts of Electricity contd.

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- **Voltage:** it can be defined as, it is the difference in charge between two points
- It means one point has more charge than other
- Unit of voltage is Volts
- It can be AC or DC
- Common example of voltage sources is battery cells which are available in market

# Concepts of Electricity contd.

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- **Current:** The amount of charge flowing through a conductor in a given time is called current
- Unit of current is **Ampere** which can be defined as one coulomb charge flowing in one second
- Symbol for representation of current is "***I***"

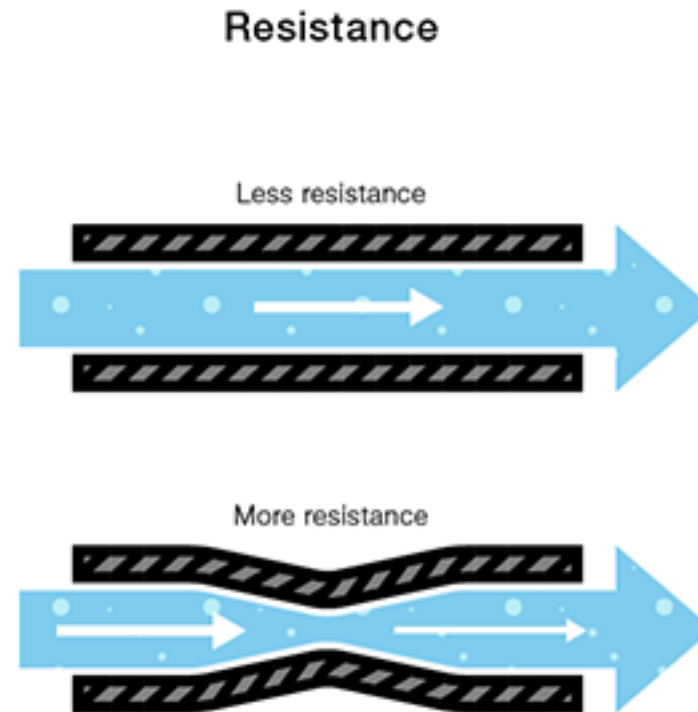
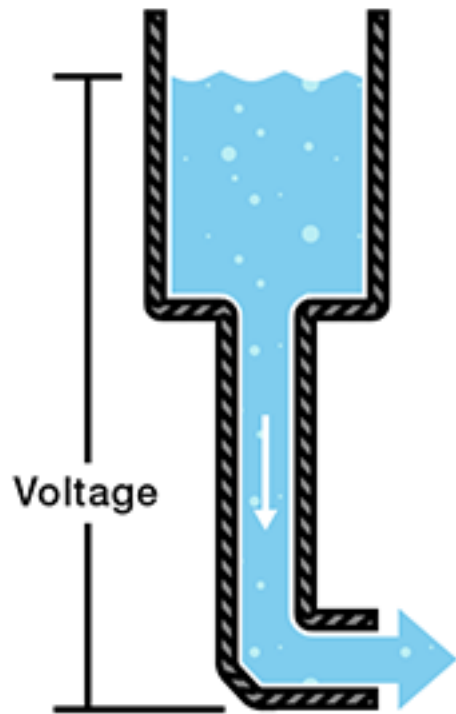
# Concepts of Electricity contd.

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- **Resistance:** Resistance is a measure of the opposition to current flow in an electrical circuit
- Unit of Resistance is **Ohm** and can be described as when a constant potential difference of one volt, applied to two points, produces in the conductor a current of one ampere
- Symbol of resistance is “**R**”

# Concepts of Electricity contd.

- All these three principles are well explained via water tank philosophy



# Ohm's Law

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- Ohm's Law establishes a relationship between voltage and current through a resistance
- This relationship established as

$$V = I \times R$$

- This is a linear equation means the plot between voltage and current will be a straight line when resistance is constant



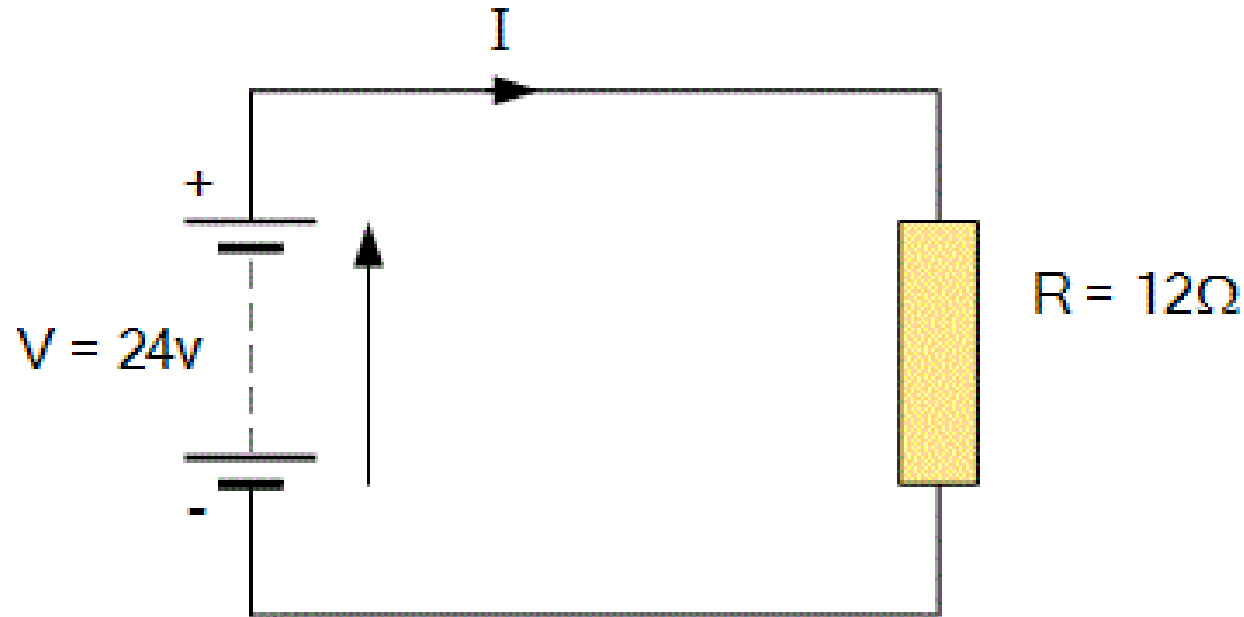
# Ohm's Law contd.

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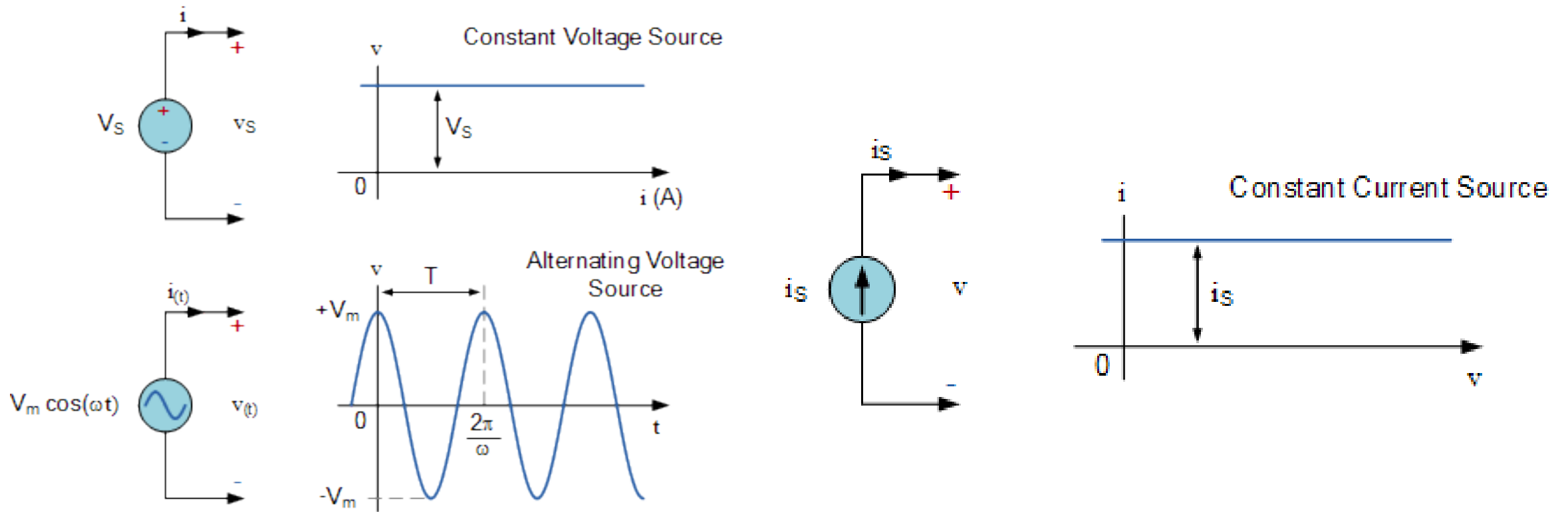
- **Electrical Power (P)** in a circuit is the rate at which energy is absorbed or produced within a circuit
- A source of energy such as a voltage will produce or deliver power while the connected load absorbs it
- Mathematically we can write Power as
$$P = V \times I$$
- The **Units of Power is Watt (W)**, milliwatt (mW) or kilowatt (KW) is also use extensively in electronics and electrical circuits

# Ohm's Law contd.

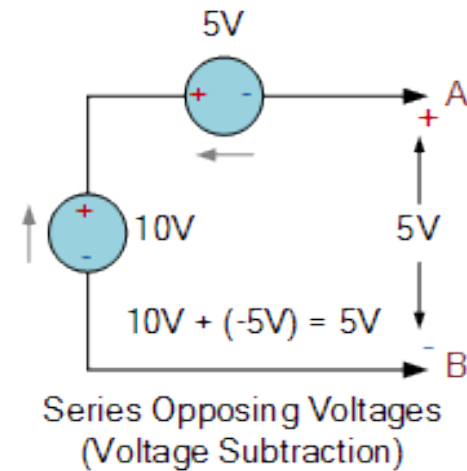
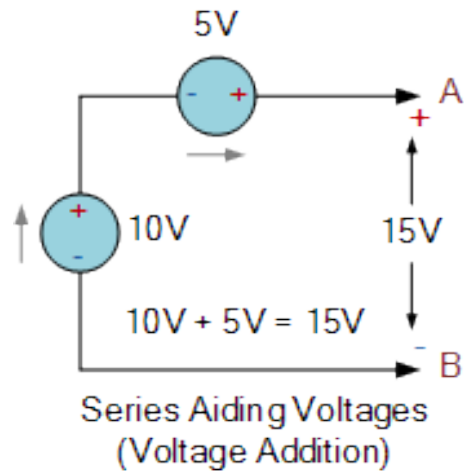
- For the circuit shown below find the Voltage (V), the Current (I), the Resistance (R) and the Power (P)



# Voltage and Current Sources



# Voltage and Current Sources contd.



$V_S =$

